

Medallion architecture plan

Bronze:

Files:

1. Airport flight data (CSV files, EuroControl, <https://ansperformance.eu/data/>)
airport_flights_(2020-2025)_raw.csv (5 files)
2. Airport locations (mapping, <https://openflights.org/>)
airport_locations_raw.csv

Information:

Scope: European airports, 5 years (2021-2025). Add columns: source name, file/API name, load timestamp or run id. Source data quality checks.

Silver:

Tables:

1. Airport flight table: Combining airport yearly data (concat)
stg_airport_flights
2. Airport geographics
airport_geography
3. Weather data (Openmeteo weather: <https://open-meteo.com/>) (airport location information: from Airport locations)
stg_weather_daily

Validations:

(silver tables joined)

4. validated_rows
5. rejected_rows

Information:

Silver must contain cleaned, standardized (columns), typed and validated data. This is where column names, dates, numeric values, category names and join keys should become reliable. Quality checks: (duplicates, null values, valid values, valid range, summary for rejections reasons).

Gold:

Tables:

fact_flights

dim_date

dim_geography

Information:

Grain: flights.

PowerBI

Analyze European airport flight traffic from 2021–2025 and evaluate how weather conditions relate to daily flight volumes across airports, countries, and seasons.